PEDRAM MOHAMMADI

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PROFESSIONAL SUMMARY

- Expert knowledge and understanding in the areas of:
  - Subjective and objective image/video quality assessment, Computational photography, Statistical analysis, Human Visual System (HVS), Color processing, Video compression, Image/Video signal processing and algorithm development
- Good programming skills using Python, MATLAB, C, and C++
- Familiar with OpenCV, Scikit-learn, FFMPEG, Linux, Git, Davinci Resolve, Elecard StreamEye

EDUCATION

Ph.D. in Electrical and Computer Engineering *2015 - 2020*
University of British Columbia, Canada

- Thesis: High Dynamic Range (HDR) video processing
- Awards: Full graduate scholarship, International Tuition Award

M.A.Sc. in Electrical Engineering *2012 - 2014*
Ferdowsi University of Mashhad, Iran

- Thesis: Subjective and Objective Image and Video Quality Assessment

PROFESSIONAL EXPERIENCE

Senior Video Quality Engineer *2021 - Present*
NETINT Technologies Inc., Canada

- Designed and developed patented video processing algorithms for enhancing the visual quality of NETINT's next-generation VPUs resulting in an average of 15% increase in objective quality.
- Lead the visual quality enhancement team in implementing various patented algorithms that enhance compression efficiency of NETINT's next-generation VPUs by an average of 10%.
- Designed and developed a patented real-time SDR/HDR conversion suite for NETINT's next-generation VPUs.
- Conducted extensive fine-tuning experiments for various company product lines resulting in an average of 5% increase in objective quality.
- Mentored junior engineers and co-op students, providing ongoing support to meet project goals.

Scientific Research and Experimental Development (SR&ED) Consultant *2020 - 2021*
KPMG LLP, Canada

- Successfully increased SR&ED tax credits for clients by an average of 20-30% annually, resulting in a total of 500K to 2 million dollars in additional credits and refunds over 1.5 years.
- Improved client cash flow by facilitating the timely receipt of SR&ED refunds, contributing to a 10-15% increase in reinvestment funds for R&D activities.

Research Scientist *2015 - 2020*
The University of British Columbia, Canada

- Designed and patented an SDR-to-HDR conversion algorithm, enhancing subjective visual quality by an average of 80%.
- Devised an image segmentation algorithm based on entropy theory capable of dividing a frame to various brightness regions.
- Developed a color adjustment algorithm, improving color accuracy by and average of 70%.
- Designed a flickering reduction algorithm for visual stability.
- Authored multiple research papers in prestigious IEEE journals, and conferences.

Application Developer I *2018 - 2019*
TELUS Communications Inc., Canada

- Led a team of 3 engineers to implement an SDR to HDR conversion system, achieving real-time performance for 1080p content.
- Developed high-quality code, enhancing product performance and user experience.

PATENTS

- **P. Mohammadi**, J. Li, E. Andrade Neto, and J. Lou "Methods and Apparatus for video rate control," Patent filed, filing date: January 2024.
- **P. Mohammadi**, E. Andrade Neto, and J. Lou "Image processing methods, devices, electronic devices and storage media," China Patent Office, Application No. CN116167950B, Publication date: April 2023.
- P. Nasiopoulos, S. Ploumis, M. T. Pourazad, R. Boitar, and **P. Mohammadi**, "Methods and Apparatuses for Tone Mapping and Inverse Tone Mapping," U.S. Patent, Application No. US11100888B2, Publication date: January 2019.

PUBLICATIONS

For a complete list of my publications please visit: My Google scholar page

WORK AUTHORIZATION

Currently holding a Canadian citizenship